TAHOSSA NWZ MOCK EXAM AUGUST 2025
B/MATHEMATICS

MARKING SCHEME

1. (a) Write a number in which O represents hundreds, 7 represents ten thousands, 3 represents ones, 9 represents thousands, 6 represents hundred thousand and 8 represents tens. Solution

Solution

- 0 –hundreds
- $7 - ten$ thousands
- 3 –ones
- 9- thousand ➢ 6 – hundred thousands
- 8 – ten
- ∴ 679083 ……02 marks

(b) Convert $\frac{125}{100}$ and 360% to decimal, then approximate their products to one significant figure.

Solution

$\boxed{\frac{125}{100} = 1.25}$

$\boxed{\frac{360}{100} = 3.6}$

⇨ ……0.5 mark

⇨ ……0.5 mark

⇨ $\frac{125}{100} \times \frac{360}{100}$

⇨ $\frac{5}{4} \times \frac{18}{5} = \frac{9}{2} = 4.5 \approx 4$ ……

…01 mark

(c) $14 - [-2 - (8 \div 2) + 5]$ solution

⇨ $14 - [-2 - 4 + 5]$

⇨ $14 - -1$

⇨ 15……02 marks

2. (a) If $\log 2 = 0.3010$, Find the value of $\log 5$ Solution

⇨ Given that $\log 2 = 0.3010$

⇨ $\log 5 = \log \frac{10}{2}$

$= \log 10 - \log 2$

$= 1 - 0.3010$ .....

……

01 marks

⇨

⇨ 0.5 marks

⇨ =0.6990 …… 0.5 marks

(b)If $(3^{x+1})(2^{2y+9}) = 69984$ find the values of x and y.

Solution

$3^{x+1} \times 2^2$ ⇨ $7^{y+9} = 3^7 \times 2^5$

⇨ $3^{x+1} = 3$

⇨ $x + 1 = 7$

⇨ $x = 6$ and $2^{2y+9} =$

$2y + 9 = 5$

$2y = -4$

…… ……01

⇨ 2^5

⇨

⇨

⇨ $y = -2$ ……01

⇨ $(x, y) = (6, -2)$

(c)Simplify by rationalizing the denominator of $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{5}+\sqrt{2}}$

Solution

⇨ $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{5}+\sqrt{2}} \times \frac{(\sqrt{5}-\sqrt{2})}{(\sqrt{5}-\sqrt{2})}$……01

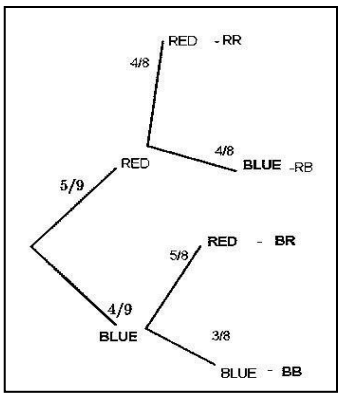
⇨ $\frac{\sqrt{15}+\sqrt{10}-\sqrt{6}-2}{5-2}$

⇨ $\frac{\sqrt{15}+\sqrt{10}-\sqrt{6}-2}{3}$ ……01

3. (a) A bag contains 5 red and 4 blue balls. Two balls are drawn in succession without replacement. Find the probability that both are red.

Solution

⇨ Sample space = 5r+4b= 9



– – —

.....01 mark Tree diagram

⇨$P(both\ R) = P(R) \times P(R)$.....01 mark

⇨$P(RR) = \frac{5}{9} \times \frac{4}{8} = \frac{20}{72} = \frac{5}{18}$.....01 mark

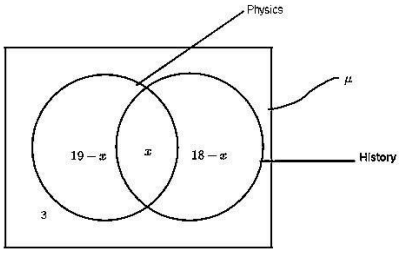
(b) In a class of 35 students, 19 take physics, 18 take History and 3 take neither of the two subjects. Calculate how many students take i) Both subjects ii) Physics or History Solution

⇨ $n(\mu) = 35$

⇨ $n(p) = 19$

⇨ $n(h) = 18$⇨ $n(p \cup h)' = 3$

⇨ Venn diagram



⇨ $19 - x + x + 18 - x + 3 = 35$

⇨ $-x + 40 = 35$

⇨ $x = 5$

⇨ (i) Both is 5 students

.....01 mark

⇨ (ii) Physics or history is 35-5 =30 students .....02 marks

4. (a) Find the equation of a line passes through the point of intersection of the lines 3x -2y -8=0 and 2x+y=3 and is perpendicular to line 4x +3y=0

Solution

⇨ Solve the equation $\begin{cases} 3x - 2y = 8 \\ 2x + y = 3 \end{cases}$

⇨ $(x, y) = (2, -1)$

⇨ Now $4x + 3y = 0 \perp (2, -1)$ ⇨ $3y = -4x$

⇨ $y = -\frac{4}{3}$

$for \perp = M_1M_2 = -$ ⇨ 1

⇨ $-\frac{4}{3}m_2 = -1$

$m^2 = \frac{3}{4}$

⇨ ……02 marks

⇨ $eq: y = m(x - x_1) + y_1$

⇨ $y = \frac{3}{4}(x - 2) - 1$

⇨ $4y = 3x - 6 - 4$

⇨ $3x - 4y - 10 = 0$……01 mark

(b) If **a** =3j – i and **b**=5i-2j and **c** = 4a + 3b find

(i).Magnitude of **ç**

(ii).Direction of **ç** Solution

(i) Magnitude $= |c| = \sqrt{x^2 + y^2}$ ……0.5 mark

⇨ $c = 4(3j - i) + 3(5i - 2j)$

⇨ $12j - 4i + 15i - 6j$

⇨ $11i + 6j$……01mark

⇨ $|c| = \sqrt{11^2 + 6^2} = \sqrt{121 + 36} = \sqrt{(157)}$

(ii) direction

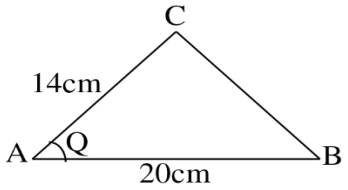
⇨ $tan\theta = -\frac{y}{x}$……0.5 mark

⇨ $\theta = \tan^{-1}(\frac{6}{11}) = 28^0 37'$…… ……01 mark

5. (a) Given that Area =$140cm^2$

Length$\bar{AB} = 12cm$

$\bar{AC} = 14cm$



Area =$\frac{1}{2}AB \times AC sin\theta$……1marks

$140 = \frac{1}{2} \times 20 \times 14\ sin\theta$

$140 = 10 \times 14\ sin\theta$

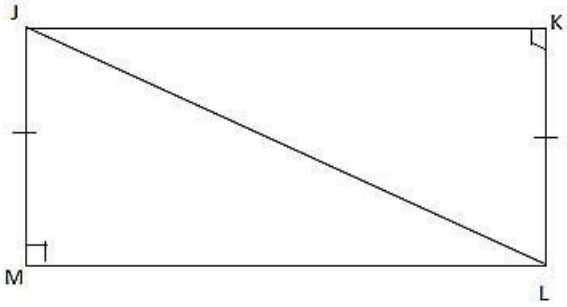
$\frac{140}{140} = \frac{140}{140}\ sin\theta$

$1 = sin\theta$

$\theta = sin^{-1}(1)$

$\theta = 90^o$ ……02 marks

(b)(i)From the Rectangle JKLM given below (i) name two triangles which are congruent. (ii) Prove your answer in (i) above



Solution

- ⇨ $MJL \approx JLK$……01 mark
- ⇨ $JML \approx JKL$……01 mark

(ii) Angle MJL is equal to JLK by SSA……01 mark

6. (a) Given that (Z + 1) is directly proportional to x and inversely proportional to the square root of y. If x =2 when y=4 and Z =4 find Z when X = 3 and y =9

Solution

- ⇨ $(Z+1) \propto \frac{x}{\sqrt{y}}$
- ⇨ $Z+1 = \frac{kx}{\sqrt{y}}$
- ⇨ $5 = \frac{2}{2}k$
- ⇨ $k = 5$ Then
- ⇨ $Z+1 = \frac{(5\times3)}{\sqrt{9}}$ ……01 mark
- ⇨ Z+1$= \frac{15}{3}$
- ⇨ $z+1=5$
- ⇨ $z=4$……02 marks

(b) A bus travels 240 km using 16 liters of diesel. How many liters of diesel are needed to drive 90km?

Solution

- ⇨ $240km = 16l$⇨ $90km$ =?
- ⇨ $240l = 90 \times 16$
- ⇨ $l = 6$

=6 litres……03 marks

7(a) On January 1st 2015 MTL group started a business with a capital in cash of 2,000,000/=

January 2. Purchased goods for cash 1,400,000/=

3. Sold good for cash 1,000,000/=

6. Purchased goods for cash 60,000/=

15. Paid rent for cash 220,000/=

26. Sold goods for cash 620,000/=

Prepare the cash account and balance it

Solution

MTL GROUP JANUARY 2015

Dr				Cr			
Date	Particular	Folio	Amount	Date	Particular	Folio	Amount
1/1	Capital	01	2,000,000/=	2/1	Goods	02	1,400,000/=
3/1	Sale	03	1,000,000/=	4/1	Goods	04	600,000/=
26/1	Sale	06	620,000/=	15/1	Rent	05	220,000/=
						b/c	
				31/1	Balance		1400000/=
			3,620,000/=				3,620,000/=
01/02	Balance	d/c	1,400,000/=				

ALL TABLE ……03 marks

b) If the ratio of sacks of maize, millet and cassava in certain store is 6:7:3. There are 42
sacks of millet. Find the total sacks in a store.

Solution

- ⇨ Total ratio = 6 + 7 + 3 = 16……01 mark
- ⇨ Total store = x
- ⇨ For maize $\frac{6}{16} \times x$……01mark
- ⇨ $\frac{6}{16} \times x = 42$
- ⇨ $6x = 42 \times 16$
- ⇨ $x = \frac{42 \times 16}{6} = 112$
- ⇨ Total sacks is 112……01 mark

8(a) If the common difference of an A.P is -2 and the third term is 0. Find

(i) The first term

(ii)The general term Solution

(i) $A_n = A_1 + (n-1)$

- ⇨ $A_3 = A_1 + (3-1) - 2$
- ⇨ $0 = A_1 - 4$
- ⇨ $A_1 = 4$
- ⇨ First term is 4……1.5 marks

(ii) General term

- ⇨ $A_n = A_1 + (n-1)d$
- ⇨ $A_n = 4 + (n-1) - 2$
- ⇨ $A_n = 4 - 2n + 2$
- ⇨ $A_n = 6 - 2n$……1.5 marks

(b) A certain geometric progression has a common ratio of 2 and the sum of the first ten terms is 155. Find the first term and give the formula for the nth term Solution

⇨ $S_n = \frac{G1(r^n-1)}{r-1}$

⇨ $155 = \frac{G1(2^{10}-1)}{2-1}$

⇨ $155 = 1023G_1$

⇨ $G_1 = \frac{155}{1023}$……1.5 marks

⇨ First term is $\frac{155}{1023}$

⇨ The $nth = G_n = G_1 r^{n-1}$

⇨ $G_n = \frac{155}{1023} \times 2^{n-1}$ ……1.5 marks

9(a) Without using tables, find the value of $\frac{\sin 60^0 \cos 60^0}{\tan 60^0}$ and simplify your answer.

Solution

⇨ $\sin 60^0 = \frac{\sqrt{3}}{2}$

⇨ $\cos 60 = \frac{1}{2}$

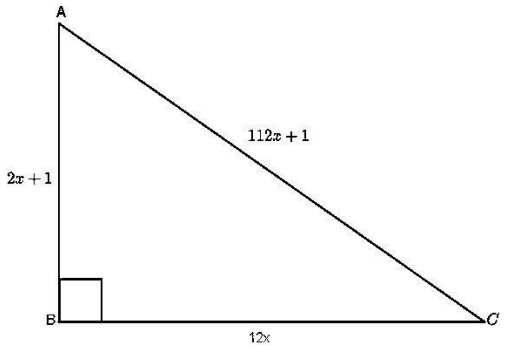
⇨ $tan60° = \sqrt{3}$

⇨ $\frac{\sin 60^0 \cos 60^0}{\tan 60^0} = \frac{\frac{\sqrt{3}}{2} \times \frac{1}{2}}{\sqrt{3}}$ ……02 marks

⇨ $\frac{\sin 60^0 \cos 60^0}{\tan 60^0} = \frac{\sqrt{3}}{4} \times \frac{1}{\sqrt{3}} = \frac{1}{4}$ ……01 mark

(b)The sides of a right angle are $2x + 1, 12x$ and $12x + 1$. Find the value of $x$in cm if the hypotenuse is $12x + 1$

Solution



⇨ $(12x)^2 + (2x + 1)^2 = (12x + 1)^2$ ……01 mark

⇨ $144x^2 + 4x^2 + 4x + 1 = 144x^2 + 24x + 1$…… ……01 mark

⇨ $4x^2 = 20x$
⇨ $x = 5cm \; or \; x = 0$……01 mark

10(a) Use factorization method to find the value of X if $8x^2 - 34x + 21 = 0$
Solution

⇨ $8x^2 - 34x + 21 = 0$
⇨ Sum $= -34$
⇨ Product $= 168$
⇨ Numbers $= (-24, -7)$……01 mark ⇨
$(8x^2 - 24x)(-7x + 21) = 0$……01 mark
⇨ $8x(x - 3) - 7(x - 3) = 0$
⇨ $(8x - 7)(x - 3) = 0$
⇨ $8x - 7 = 0 \; or \; x - 3 = 0$
⇨ $x = \frac{7}{8} \; or \; x = 3$ ……01 mark

(b) The average of two numbers is 7 and three times the difference between them is 18.

Find the two numbers Solution

⇨ $\frac{x+y}{2} = 7$……01 mark
⇨ $3(x - y) = 18$
⇨ $x + y = 14$……01 mark
⇨ $x - y = 6$
⇨ Solve simultaneous equation
⇨ $(x, y) = (10,4)$

The numbers are 4 and 10……01 mark

SECTION B (40 MARKS)

11(a)These scores of Mathematics Examination done by 60 students were recover as follow

36,**26**,**52**,**54,51**,**34**,**26**,**29**,**34**,**26**,**30**,**56**,**21**,**49,35**,**5822**,**38**,**28**,**26**,**41**,**53**,**31**,
32,**20**,**21**,61,**58,36**,**50**,**56**,63
27,**30**,63,**30**,60,**34**,**48**,**28**,**41**,**57**,**24**,**35,35**,**33**,**20**,**59**
36,**25**,**32**,**43**,**56,30**,**50**,**42**,**21**,**33**,**45**,**25**

i) Prepare frequency distribution table using intervals 20- 24,25 -29, 30-34 ii) Calculate mode and Median

⇨ Solution ……03 marks

Class interval	Class mark	Frequency	Tally	C. F

20-24	22	8	///////	8
25-29	27	10	////////	18
30-34	32	12	//////////	30
35-39	37	7	//////	37
40-44	42	4	////	41
45-49	47	3	///	44
50-54	52	6	/////	50
55-59	57	7	//////	57
60-64	62	4	//	60

(i) $Mode = L + \left(\frac{t_1}{t_1+t_2}\right)C$

$29.5 + \left(\frac{2}{2+5}\right) \times 5$

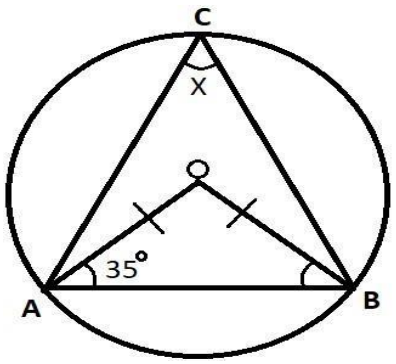
⇨ $29.5 + \frac{10}{7} = 30.9$ ……1.5 marks

(ii) $Median = L + \frac{(N-n_b)}{f_m}C$

$= 29.5 + \frac{(30-18)}{12} \times 5$

⇨ $=29.5 + \frac{60}{12} = 33.5$ ……1.5 marks

(b) solution



ΔAOB is an isosceles

OA=OB= r (radii)

$\Delta AOB = \Delta OBA$ (Base angle of an isosceles)……01mark

Δ

Thus $+ \Delta AOB = 180^0$

But $\Delta AOB = 2\Delta ACB$.....01 mark

$\Delta ACB = x$

$\Delta AOB = 2x$

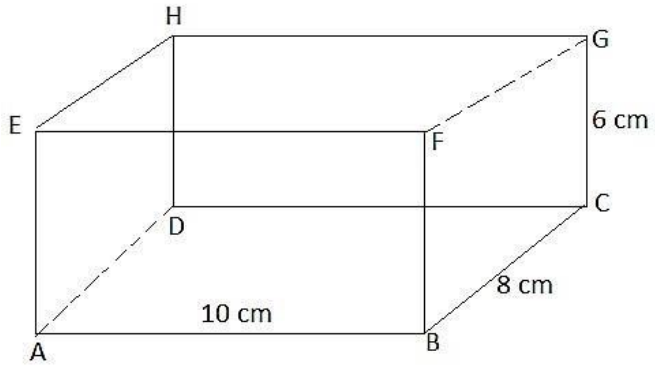
$\frac{110}{2} = \frac{2x}{2}$

$x = 55^0$

Therefore the value of $x = 55^o$ ..... 01 mark

12 (a) The figure below shows a rectangular prism in which AB= 10CM, BC= 8cm and GC =
6cm

(i).Calculate the length AG (Two decimal places)



(ii).The angle that AG makes with the plane ABCD

Solution

AG is under ΔAGC

But $GC = 6d\ AC$ is the diagonal of $ABCD$

Use $(AB)^2 + (BC)^2 = (AC)^2$ .....01 mark

$10^2 + 10^2 = (AC)^2$

$(AC)^2 = 200$

$AC = 10\sqrt{2}$ .....01 mark

Now use $(AC)^2 + (GC)^2 = (AG)^2$

$200 + 6^2 = AG^2$

$236 = AG^2$

AG = $\sqrt{236}$

cm .....01 mark

(ii) Angle AG

$tan\theta = \frac{CG}{AC} = \frac{6}{10\sqrt{2}}$.....01 mark

$\theta = tan^{-1}\left(\frac{6}{10\sqrt{2}}\right) = 22.99^0$ .....01 mark

(b) A and B are two points on latitude 70°N. Their longitudes are 62°W and 118°E respectively. Calculate the distance in Kilometers from A to B if the Earth's diameter is 12,800km along a parallel of latitude Solution

Formula $= \frac{\theta}{360}{}^{0} 2\pi R cos\alpha$ .....01 mark

$= \frac{62+118}{360} \times 2 \times 3.14 \times 6400 \times \cos 70^0$ ……02 marks

$=$6873.3km ……02 marks

13(a) I think two numbers. If I double the first and add the second I get 18. If I double the first and subtract the second I get 14. Use matrix method to find the two numbers

Solution

$2x + y = 18$

$2x - y = 14$

$\begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 18 \\ 14 \end{pmatrix}$……01 mark

$Let\ A = \begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix}$

$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} -1 & -1 \\ -2 & 2 \end{pmatrix}$

$A^{-1} = \frac{1}{-4}\begin{pmatrix} -1 & -1 \\ -2 & 2 \end{pmatrix}$

$A^{-1} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{-1}{2} \end{pmatrix}$……01 mark

$\begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{-1}{2} \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{-1}{2} \end{pmatrix}\begin{pmatrix} 18 \\ 14 \end{pmatrix}$……01 mark

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{32}{4} \\ \frac{12}{4} \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 3 \end{pmatrix}$

Thenumbers are 8 and 3 ……01 mark

(b) Given $\begin{pmatrix} 2t+2 & t \\ 4t-3 & t+3 \end{pmatrix}$ is singular

$|\Delta| = 0$

$(t+3)($

$(2\ t+2) + 3(2t+2) - 4t^2 + 3t = 0 \quad 2t+2) - (4t-3) = 0$

$2\ t^2 + 2t + 6t + 6 - \qquad t = 0 \quad 4t^2 + 3$

…… …01

$2t^2 + 8t + 3t + 6 = 0$

$t^2 + 11t + 6 = 0$

-2

$2t^2 - 11t - 6 = 0$ ……01

$2t^2 + t - 12 \quad t - 6 = 0$

or t-6=0 $(2t + 1) - 6(2 \quad t \quad + 1) = 0$

$\frac{2t}{2} = \frac{-1}{2}$ or $(2t + 1)(t - 6 \quad) \quad = 0$

$2t + 1 = 0$

t=6 $t = {}^{-1}/_{2}$ or t=6

Therefore t=$-1/2$ *or* 6 …….. 01 mark

(c) Find the image equation of the line $y = 3x + 7$ after reflection in the line $y = x$
Solution

$y = x$, its matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$……01 mark

And y=3x+7 use x=0 , y=7 and x=1 ,y=10

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 7 \end{pmatrix}=\begin{pmatrix} 7 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 10 \end{pmatrix}=\begin{pmatrix} 10 \\ 1 \end{pmatrix}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-1}{7-10} = \frac{1}{3}$ ……01 mark

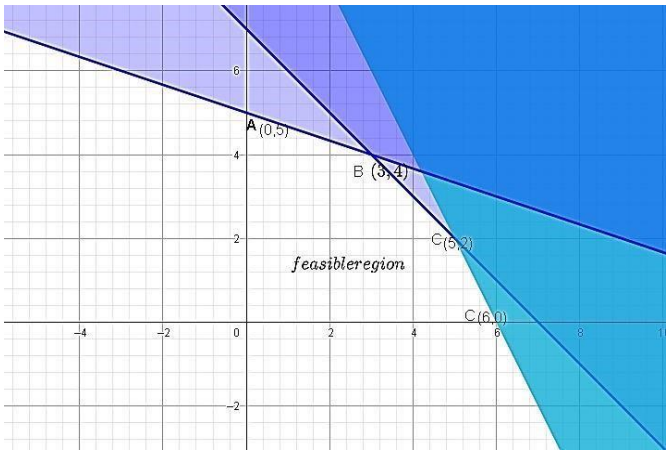
$y - 1 = \frac{1}{3}(x - 10)$

$\therefore 3y = x - 7$……01mark

14(a) Maximize $(x, y) = 2x + 6y$ subject to the following constraints

$x + \leq 7$
$2x + \leq 12$ $x +$
$3y \leq 15$
$\geq 0, y \geq 0$

Graph



……03

Table result

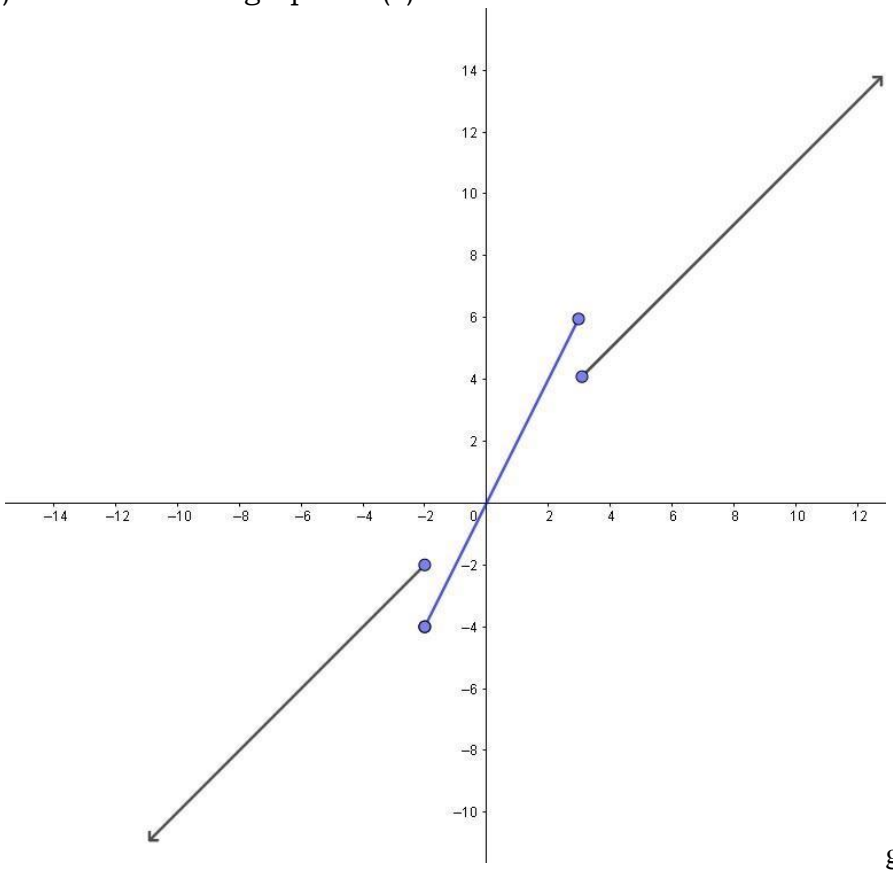
Corner point	$f(x, y) = 2x + 6y$	Total value

A(0,5)		30
B(3,4)		30
C(5,2)		22
D(6,0)		12

The maximum is 30 ……02

(b) The function f(x) defined as:- f(x) = $\begin{cases} x\ if\ x < -2 \\ 2x\ if\ -2 \leq x < 3 \\ (x+1) if\ x \geq 3 \end{cases}$

(i) Sketch the graph of f(x)



graph……03 marks

(ii) State domain and range of f(x)

Domain

$= x: x \in IR$……01

Range

$= y: y \in IR$……01